⚖️ KanoonSaathi — India's AI-Powered Legal Companion

> **"Know Your Rights. Speak the Law."** Free legal knowledge for every Indian citizen — no lawyer needed.

📌 What Is This App?

KanoonSaathi (meaning *Legal Companion* in Hindi) is an AI-powered web application built to make Indian law accessible to every citizen — including those who are uneducated, cannot afford a lawyer, or have never interacted with the legal system.

India has over 1.4 billion people. 3 out of 4 Indians do not know what laws protect them. Most cannot afford a lawyer. KanoonSaathi bridges that gap by putting the entire Indian legal system in their hands — free, instant, and in their own language.

Built for **The Big Code 2026 Hackathon** as a solo project.

🎯 Who Is It For?

User	How It Helps
Uneducated / Illiterate users	Speak their problem via mic — get an audio response back
Workers & labourers	Know their wage, PF, gratuity and workplace rights
Women	Access safety apps, domestic violence laws, POSH Act
Farmers & rural citizens	Understand land rights, RERA, consumer protection
Students	RTE Act, anti-ragging, UGC fee refund rules
Commuters & travellers	Flight/train rights, DGCA rules, TTC misconduct
Persons with disabilities	RPwD Act rights, UDID card, accessible transport
Anyone facing police / court	FIR rights, arrest rights, free legal aid

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🚀 Core Features

1. 🤖 AI Legal Assistant (Chat)

- Ask any legal question in plain language
- AI responds with: applicable law + section, plain explanation, step-by-step action plan, offender's punishment, free help options, key helplines
- Supports **multi-turn conversation** within a session
- Auto-retry on rate limits (3 attempts with fallback to Haiku model)

2. 🎙️ Voice Mode (Full Speak-to-Speak)

- **Microphone input** — tap mic, speak your problem, AI answers automatically
- **Text-to-speech output** — AI response read aloud after every reply
- Live sound wave animation while listening
- Live transcript shown as you speak
- Auto-submits query when speech ends — no typing needed
- 🔊 / 🔇 toggle to turn voice replies on/off
- Manual replay button on every AI message
- 🔊 button on Daily Law Fact to hear it aloud
- Covers all 23 languages with correct BCP-47 speech codes
- Proper microphone permission request with user-friendly error messages

3. 🌐 23 Languages

All 22 languages listed in the **8th Schedule of the Constitution of India** + English:

Hindi, Tamil, Telugu, Kannada, Malayalam, Bengali, Marathi, Gujarati, Punjabi, Odia, Assamese, Urdu, Nepali, Kashmiri, Konkani, Maithili, Dogri, Manipuri, Santali (Ol Chiki script), Bodo, Sindhi, Sanskrit + English

- Full UI translation (hero text, tabs, buttons, chat placeholder)
- RTL layout for Urdu, Kashmiri, Sindhi
- Native script display for all languages
- AI instructed to respond in the selected language
- Language picker with search by name, native script, or region

4. 📚 20 Law Categories (180+ Sections)

Category	Key Laws
👥 Common Citizens	Art. 14, 19, 21; IPC 302, 379, 420, 498A, 500; IT Act 66C/66D
🛒 Consumer Rights	CPA 2019, GST Rules, E-Commerce Rules, LMA 2009
✈️ Tourist & Travel	DGCA CAR, Montreal Convention, IPC 354/509
🏛️ Govt Employees	RTI Act, CCS Rules, Whistleblower Protection Act, Art. 311
💼 Private Employees	MWA, Gratuity Act, EPF, POSH Act, Maternity Benefit Act
🍱 Food Safety	FSS Act Sec 59/63, FSSAI Reg, LMA 2009
🏠 Real Estate (RERA)	RERA Sec 3/12/14/18, IPC 420
💰 Tax Laws	ITA Sec 80C/80D/194/276C, GST Sec 171
📋 RTI Act	RTI Sec 3/6/7/8/20
🐾 Animal Rights	PCA Act, ABC Rules 2001, WPA 1972
🏥 Medical Laws	IPC 304A, CPA, CEA 2010, D&C Act, THO Act
🚔 FIR & Police Rights	CrPC 154/41/41A, Art. 22, D.K. Basu, IPC 166
🏗️ Labour Laws	MWA, EPF, ESI, IDA, Factories Act, BOCW, Child Labour, Trade Unions, Labour Codes, eShram
💸 ED & PMLA	PMLA Sec 3/5/19/45/50, FEMA, OSA
🎓 Education Laws	RTE Act, UGC Anti-Ragging, UGC Refund Policy, RPwD, POCSO
🔍 CBI	DSPE Act 1946, PC Act 1988, CBI jurisdiction
🚚 ARAI & Automotive	MVA 1988, BS6 norms, ARAI homologation, MACT
🎖️ Armed Forces	Army/Navy/Air Force Acts, AFSPA, Official Secrets Act, IPC 131/353
🛡️ BSF & Border Laws	BSF Act 1968, ILP Rules, BADP, Customs Act

Category	Key Laws
🚂 Railways	Railway Act Sec 137/138/142/145/124, PC Act (TTC bribery), RailMadad

5. 🎭 14 Common Scenarios

Expandable cards with: situation description, applicable laws, numbered action steps, offender punishment, warning box, helpline numbers, and "Get AI Advice" button.

Covers: Phone theft, Builder fraud, Food poisoning, Flight cancellation, Salary not paid, Online scam, RTI filing, Medical negligence, Police FIR refusal, Defective product, Dowry harassment, Tax notice, Hotel/GST overcharging, TTC misconduct in trains.

6. 🚨 9 What If Situations

Emergency action guides for real-time crises:

- Someone threatens you (IPC 503/506)
- You are being ragged (UGC + IPC 323/294)
- Weapon threat on road (IPC 397 / Arms Act)
- Road accident (MVA 134 / IPC 279/304A)
- Police is arresting you (CrPC 50 / Art. 22)
- You got cyber-scammed (IT Act 66C/66D)
- What if YOU threaten someone (IPC 506 — reverse awareness)
- TTC demanding bribe in train (Railway Act + PC Act)
- Someone near you is misbehaving/creating nuisance (IPC 268/290/294/510)

7. ⚖️ Know Your Rights (12 Rights)

Fundamental rights every Indian should memorise — with the exact article, plain description, and what to say/do. Covers: right to silence, free legal aid, RTI, education, vote, peaceful protest, emergency treatment, equal pay, bail, unlawful search, speech, and knowing reason for arrest.

8. 👩 Women's Safety (10 Apps & Helplines)

Detailed cards for each resource:

- Himmat Plus (Delhi Police app), 112 India App, Mahila Helpline 181, iCall (TISS), Nirbhaya App, NCW (WhatsApp: 7217735372), One Stop Centre (Sakhi), Sheroes App, Ola/Uber SOS, POCSO e-Box
- Quick-dial strip with 5 emergency numbers
- What it does, how to use it, who it's useful for

9. ♿ Disability Rights (10 Sections)

Rights under RPwD Act 2016:

- Equal employment, inclusive education, 4% job reservation
- Accessible transport, free medicines, disability pension
- UDID card (swavlambancard.gov.in), protection from abuse
- Mental illness rights under Mental Healthcare Act 2017
- Helpline: 1800-11-4515

10. ⭐ Bookmarks

- Star (☆/⭐) button on every law card and scenario card
- Persisted in **localStorage** — survives page refresh
- Dedicated **Bookmarks tab** with grouped sections (Laws / Scenarios)
- Ask AI directly from any bookmarked item
- View Category / View Details buttons
- Clear All option
- Live count shown in tab label

11. 📅 Daily Law Fact

- Rotates daily (30 unique legal facts)
- Covers everything from PF to AFSPA to railway laws
- 🔊 button to hear the fact read aloud
- Shows exact law citation

12. 🔐 Authentication (Login & Signup)

- Full-screen Login page with email + password
- Full-screen Signup page with name, email, phone, password, confirm password
- Live password strength bar (Weak / Medium / Strong)
- Real-time passwords match indicator
- Phone number auto-strips non-digits
- User accounts stored in sessionStorage (demo — ready for real backend)
- "Continue as Guest" option

13. 👤 User Profile

- Editable: Date of Birth, Gender, City, State (all 36 states/UTs), Occupation, Preferred Language
- Read-only: Name, Email, Phone (from account)
- Auto-generated initials avatar
- DPDP Act 2023 privacy note
- Save confirmation animation

14. 📞 19 Emergency Helplines

Displayed as a card grid with name, number, description. Includes: Police (100), Emergency (112), Ambulance (108), Fire (101), Mahila (181), Child (1098), Cyber Crime (1930), Consumer (1915), FSSAI (1800-112-100), DGCA, Labour (14567), Bank Ombudsman (14448), Anti-Corruption (1064), DLSA (15100), Senior Citizen, Mental Health (iCall), Railway (139), GRP (1512), RailMadad portal.

🗂️ App Structure

```
KanoonSaathi/
│
├── Data Layer (constants — top of file)
│   ├── SCENARIOS[]          — 14 common legal scenario cards
│   ├── HELPLINES[]          — 19 emergency helpline entries
│   ├── WOMEN_APPS[]         — 10 women's safety apps/helplines
│   ├── DISABILITY_LAWS[]    — 10 RPwD Act rights
│   ├── LAW_DB{}             — 20 categories × 141+ law entries
│   ├── WHAT_IF[]            — 9 emergency situation guides
│   ├── RIGHTS[]             — 12 fundamental rights cards
│   ├── LAW_FACTS[]          — 30 daily rotating legal facts
│   ├── LANGUAGES[]          — 23 language definitions + BCP-47 codes
│   ├── UI_TEXT{}            — Full UI translations for all 23 languages
│   ├── SPEECH_LANG{}        — Speech API locale codes for all 23 languages
│   └── SYSTEM_PROMPT        — AI instruction set covering all 20 law areas
│
├── Utility Functions
│   ├── getT(lang)           — Translates UI text for current language
│   └── cleanForSpeech(text) — Strips markdown/emoji for natural TTS
│
├── Hooks (defined outside main component)
```

```
│   └── useSpeech(lang)      — Full STT + TTS with permission handling
│
├── Components (defined outside main component)
│   ├── NavBar               — Sticky nav with language picker + auth buttons
│   ├── LanguagePicker       — Searchable 23-language modal
│   ├── Splash               — Animated loading screen with progress bar
│   ├── LoginPage            — Full-screen login with validation
│   ├── SignupPage            — Full-screen signup with strength meter
│   └── ProfilePage          — User profile editor
│
└── Main Component: KanoonSaathi()
    │
    ├── State
    │   ├── screen           — "home" | "chat" | "category" | "login" | "signup" |
"profile"
    │   ├── selCat           — selected category ID for detail screen
    │   ├── openSc           — open scenario/whatif card ID
    │   ├── messages[]       — chat history
    │   ├── loading          — API call in progress
    │   ├── loggedIn/user    — auth state
    │   ├── query            — home screen search filter
    │   ├── tab              — active home tab
    │   ├── bookmarks[]      — saved laws/scenarios (localStorage)
    │   ├── lang             — active language code
    │   ├── langOpen         — language picker modal open
    │   ├── autoSpeak        — voice reply toggle
    │   └── copied           — copy feedback state
    │
    ├── Screens (conditional renders)
    │   ├── Splash Screen    — animated loading on first mount
    │   ├── Login Screen     — full page (screen === "login")
    │   ├── Signup Screen    — full page (screen === "signup")
    │   ├── Profile Screen   — full page (screen === "profile")
    │   ├── Category Detail  — full page law list for selected category
    │   ├── Chat Screen      — full page AI conversation
    │   └── Home Screen      — main screen with all tabs
    │
    └── Home Tabs
        ├── 📚 Categories    — 20 law category cards with search filter
        ├── 🎭 Scenarios     — 14 expandable scenario cards
        ├── 🚨 What If?      — 9 emergency situation guides
        ├── ⚖️ My Rights     — 12 fundamental rights cards
        ├── 👩 Women Safety  — 10 women's safety apps/helplines
        ├── ♿ Disability    — 10 RPwD Act rights
```

```
├── ⭐ Bookmarks     — saved laws and scenarios
└── 📞 Helplines     — 19 emergency helpline cards
```

🛠️ Tech Stack

Layer	Technology
Framework	React 18 (JSX, Hooks)
Styling	Inline CSS with CSS custom properties + keyframe animations
AI	Anthropic Claude API (`claude-sonnet-4-5`, fallback `claude-haiku-4-5`)
Voice Input	Web Speech API (`SpeechRecognition` / `webkitSpeechRecognition`)
Voice Output	Web Speech API (`SpeechSynthesis`)
Storage	localStorage (bookmarks), sessionStorage (user accounts for demo)
Fonts	Georgia (serif — warm, trustworthy feel)
Icons	Unicode emoji (no icon libraries needed)
Animations	Pure CSS keyframes (fadeUp, pulse, orbit, wave, micPulse, glow)
API Endpoint	`https://api.anthropic.com/v1/messages`

✅ What Has Been Implemented

Core Features

- ☑ AI legal Q&A with structured 6-section response format
- ☑ Auto-retry on rate limits (3 attempts, exponential backoff, fallback model)
- ☑ Full voice input (STT) with permission handling and error messages
- ☑ Full voice output (TTS) with auto-speak and per-message replay
- ☑ Mic auto-submits query when speech ends
- ☑ 23 languages (22 Scheduled + English) with full UI translation
- ☑ RTL layout for Urdu, Kashmiri, Sindhi
- ☑ Language picker with search by name/script/region
- ☑ AI responds in selected language

Content

- ☑ 20 law categories covering 141+ individual laws/sections
- ☑ 14 common legal scenario cards with action steps
- ☑ 9 What If emergency situation guides
- ☑ 12 Know Your Rights fundamental rights cards
- ☑ 10 Women's safety apps & helplines with how-to-use details
- ☑ 10 Disability rights under RPwD Act 2016
- ☑ 30 Daily Law Facts (rotates daily, with TTS button)
- ☑ 19 emergency helpline cards
- ☑ GST rate table for restaurant/hotel billing disputes

User Features

- ☑ Login page with validation (email, password, show/hide)
- ☑ Signup page with password strength bar, match indicator, phone validation
- ☑ User profile with editable personal details
- ☑ Bookmarks — save/unsave laws and scenarios, persisted in localStorage
- ☑ Copy last AI response to clipboard
- ☑ Search/filter across all categories

UI/UX

- ☑ Animated splash screen with progress bar
- ☑ Orbital ring background animations
- ☑ Card hover effects with category colours
- ☑ Listening wave animation and live transcript
- ☑ Speaking banner with stop button
- ☑ Mic error banners with exact fix instructions
- ☑ Retry progress shown in loading bubble
- ☑ Auto-speak ON/OFF toggle button
- ☑ Responsive grid layout (mobile + desktop)
- ☑ Custom scrollbar styling

⬜ What Needs To Be Implemented Further

🔴 High Priority (Core Functionality)

Feature	Description
Real Database Backend	Replace sessionStorage with Firebase/Supabase. User accounts, chat history, and bookmarks should persist across devices and sessions.
OTP Verification	Real 2FA using mobile OTP (Twilio / MSG91) for Indian phone numbers. Currently signup has no OTP verification.
Forgot Password Flow	Password reset via email OTP. Currently shows "Coming soon".
Chat History	Save and reload past AI conversations per user account. Users should be able to review what they asked before.
Document Analyzer	Upload a legal notice, FIR copy, rental agreement, or employment contract. AI reads and explains it in plain language.

🟠 Medium Priority (High Impact Features)

Feature	Description
FIR Draft Generator	Based on the user's situation, AI generates a ready-to-print FIR draft in the correct format.
Legal Notice Generator	AI generates demand letters — to employer for unpaid salary, to builder for RERA complaint, etc.
PDF / Share Export	Export any AI response or law card as a PDF. Share via WhatsApp directly.
Location-Based Legal Aid	Show nearest DLSA office, consumer forum, police station based on device GPS.
State-Specific Laws	Auto-detect user's state and show state-specific minimum wages, RERA portal, consumer forum address.
Offline Mode (PWA)	Make the app installable on phone home screen. Law database works offline without internet.
Push Notifications	Send Daily Law Fact as a morning phone notification.

Feature	Description
Multi-turn AI Memory	AI should remember context within a full session — "what did I ask earlier?" should work.

🟡 Medium Priority (Content & Accuracy)

Feature	Description
Fully Translated Law Content	Currently only UI labels are translated. All 141+ law descriptions, scenario steps, and What If guides need full translation into all 23 languages.
Landmark Cases Section	Maneka Gandhi, Vishaka Guidelines, D.K. Basu, Olga Tellis — explained simply for citizens.
Legal News Feed	Daily updates on important Supreme Court and High Court judgments relevant to common citizens.
Scheme Finder	"Which government schemes am I eligible for?" — based on user's profile (state, occupation, income, disability status).
Legal Calendar / Deadlines	Limitation periods — "you have 90 days to file this complaint", "appeal within 30 days".
Know Your Documents	Explain Aadhaar, PAN, Voter ID, Passport, Ration Card, UDID — rights, uses, how to get them.
Hinglish Mode	Mixed Hindi-English support — how most North Indians actually speak and type.

🟢 Lower Priority (Growth Features)

Feature	Description
Anonymous Q&A Community	Users post questions, verified lawyers/volunteers answer publicly.
Success Stories	"I used KanoonSaathi and won my case" — motivates other users to take action.
One-Tap SOS Button	Single button: calls 112, shares GPS location, sends WhatsApp alert to 3 saved contacts simultaneously.

Feature	Description
Lawyer Referral	Connect users with verified pro-bono lawyers and legal aid organizations by district.
Case Outcome Predictor	Based on similar cases, estimate likely outcome and timeline.
Dark / Light Theme Toggle	Some users prefer light mode, especially elderly users.
Font Size Control	Larger text option for elderly or visually impaired users.
Simplified One-Tap Mode	Strip all UI — just a giant mic button. Speak → get answer. For completely illiterate users.
Dialect Support	Bhojpuri, Chhattisgarhi, Haryanvi (spoken by 200M+ but not in 8th Schedule).
Voice Language Auto-Detect	Detect which language the user is speaking and auto-switch.
Admin Dashboard	View most common queries, top searched laws, usage analytics by state/language.

📊 Current Statistics

Metric	Count
Total Lines of Code	~2,300
Law Categories	20
Individual Laws/Sections	141+
Legal Scenarios	14
What If Guides	9
Know Your Rights entries	12
Women's Safety Resources	10
Disability Rights entries	10

Metric	Count
Daily Law Facts	30
Emergency Helplines	19
Languages Supported	23
Screens / Pages	7
Home Tabs	8

🏆 Hackathon Context

Event: The Big Code 2026 **Category:** Social Impact / LegalTech / AI **Team:** Solo participant **Development:** 5-day sprint (Days 1–5) **Tech used during hackathon:** React, Claude API, Web Speech API, CSS Animations

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⚠️ Legal Disclaimer

KanoonSaathi is for **informational and educational purposes only**. It does not constitute formal legal advice. For formal legal proceedings, consult a licensed advocate or contact your nearest District Legal Services Authority (DLSA) — free legal aid helpline: **15100**.

Last updated: Day 5 of The Big Code 2026 Hackathon Built with ❤️ for 1.4 Billion Indians